الإمارات العربية المتحدة
وزارة التربية والتعليم

الميثاق الوطني للتعليم

WEEK 8 · LESSONS 3-1 · 3-2 · 3-3

# Weekly Quiz Preparation

## Grade 12 Advanced · Mathematics

كتيّب التجهّز للاختبار الأسبوعي القصير

1 2 3

| 3 | 10 | 40 | 9 |
|---|---|---|---|
| objectives<br>أهداف | marks<br>علامة | minutes<br>دقيقة | practice questions<br>تدريبًا |

Mr. Yousef Al Yousef · **Friday 23 October 2026**

**One page per objective · every solution written out step by step**

# Quiz map

3 objectives = 3 recipes · 3 أهداف = 3 وصفات

**1** **Average and instantaneous velocity** السرعة
Change in position ÷ change in time; shorter intervals approach v(a)

**2** **Sketch f′ from the graph of f** منحنى المشتقة
Level → 0 · rising → positive · falling → negative · steepest → farthest from 0

**3** **The General Power Rule** قاعدة القوة
x^r → r·x^(r−1), for any real r ≠ 0

**1 · Learn**
Read the idea and the recipe
اقرأ الفكرة والوصفة

**2 · Watch**
Cover the solution, try it, then compare
غطِّ الحل وحاول ثم قارن

**3 · Try**
Do “Your turn”, check at the back
حلّ التدريبات وصحّح

# 1 Average and instantaneous velocity
السرعة المتوسطة واللحظية

Average velocity needs two times. Instantaneous velocity is what the averages approach as the interval shrinks.

**Average velocity = (change in position) ÷ (change in time). Shorten the interval until it ends at t = a: the averages approach the instantaneous velocity.**

السرعة المتوسطة = التغير في الموضع ÷ التغير في الزمن. قصّر الفترة حتى تنتهي عند t = a فتقترب المتوسطات من السرعة اللحظية.

## Recipe الوصفة

1. **Find s at both times**
   احسب s عند الزمنين
2. **Average velocity = (s(b) − s(a)) ÷ (b − a)**
   السرعة المتوسطة
3. **Repeat on [1.9, 2], then [1.99, 2]**
   كرّر على فترات أقصر
4. **The averages approach the instantaneous velocity**
   تقترب من السرعة اللحظية

**Watch out**

- Divide by the change in time, not by the second time.
- Write the unit: feet per second (ft/s).

**Tool.** v_avg = [s(t₂) − s(t₁)] / (t₂ − t₁)

## Worked example Authored (shape of Ex 19-22)

**The position of an object is $s(t) = 2t^3 + t$ feet at time $t$ seconds. Find the average velocity between $t = 1$ and $t = 2$, and between $t = 1.9$ and $t = 2$. Then estimate the instantaneous velocity at $t = 2$.**

1. substitute t = 1 and t = 2

$$s(1) = 2 + 1 = 3, \qquad s(2) = 16 + 2 = 18$$

2. change in position ÷ change in time

$$\frac{18 - 3}{2 - 1} = 15 \text{ ft/s}$$

3. a shorter interval ending at 2

$$s(1.9) = 2(6.859) + 1.9 = 15.618, \qquad \frac{18 - 15.618}{0.1} = 23.82 \text{ ft/s}$$

4. shorter still: the averages approach the instantaneous velocity

$$[1.99, 2]: \ 24.88 \ \Rightarrow \text{ the averages approach } 25: \ v(2) \approx 25 \text{ ft/s}$$

**Your turn** · solve in your notebook · solutions at the back

1. Authored (shape of Ex 19) $s(t) = 4t^2 + 3$ feet. Find the average velocity between $t = 1$ and $t = 3$.
2. Authored (shape of Ex 20) $s(t) = t^3 - t$ feet. Find the average velocity between $t = 1.9$ and $t = 2$, and estimate the instantaneous velocity at $t = 2$.
3. Authored (shape of Ex 22) $s(t) = 5\sin(t - 1)$ feet (radians). Find the average velocity between $t = 0.99$ and $t = 1$, and estimate the instantaneous velocity at $t = 1$.

# 2 The graph of the derivative

منحنى المشتقة

You do not need a formula: read the slope of f from its graph and turn it into the height of f′.

**Where f levels out, f′ = 0. Where f rises, f′ is positive. Where f falls, f′ is negative. The steeper f is, the farther f′ is from 0.**

حيث يستوي f يكون f′ = 0، وحيث يصعد يكون f′ موجبًا، وحيث ينزل يكون f′ سالبًا، وكلما زاد الميل ابتعد f′ عن الصفر.

## Recipe الوصفة

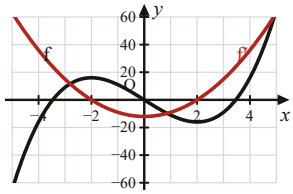
1. **Mark every point where f levels out: f′ = 0**
   علّم نقاط الاستواء: f′ = 0
2. **Rising → f′ above the axis; falling → below**
   يصعد فوق المحور، ينزل تحته
3. **Steepest places → the largest |f′|**
   أشد ميل ← أكبر |f′|
4. **Join the points with a smooth curve**
   صل النقاط بمنحنى ناعم

**Watch out**

- f′ shows the SLOPE of f, not its height.
- Do not copy the shape of f: a cubic gives a parabola.

## Worked example Book Example 2.5, p. 147

**The graph of $f(x) = x^3 - 12x$ levels out at $x = -2$ and $x = 2$. Sketch a plausible graph of $f'$.**




1. horizontal tangents: slope 0
   $f'(-2) = f'(2) = 0$ (level points)
2. positive slope: f′ above the axis
   $f$ rises for $x < -2$ and $x > 2$ : $f' > 0$
3. steepest fall at x = 0
   $f$ falls on $(-2, 2)$ : $f' < 0$, most negative at $x = 0$
4. draw the smooth curve
   $f'$ is an upward parabola through $(-2, 0), (0, -12), (2, 0)$

**Your turn** · solve in your notebook · solutions at the back

1. **Authored (shape of Ex 13)** The graph of $f(x) = x^2 - 4x + 3$ has its lowest point at $(2, -1)$. Describe the graph of $f'$: where is it zero, positive and negative, and what kind of curve is it?
2. **Authored (shape of Ex 14)** The graph of $f(x) = -x^3 + 3x$ has a high point at $x = 1$ and a low point at $x = -1$. Sketch $f'$: where is it zero, positive and negative?
3. **Authored (shape of Ex 15)** The graph of $f(x) = \frac{1}{3}x^3 - x$ levels out at $x = -1$ and $x = 1$. Describe the graph of $f'$.

# 3 The Power Rule

قاعدة القوة

One rule for every power: the exponent comes down, and the new exponent is one less.

**General Power Rule (Theorem 3.2): for any real number r ≠ 0, d/dx x^r = r·x^(r−1).**

قاعدة القوة العامة: لأي عدد حقيقي r ≠ 0، مشتقة x^r هي r·x^(r−1).

## Recipe الوصفة

1. **Write the function as x to a power**
   اكتب الدالة على الصورة x^r
2. **Bring the exponent down as a factor**
   انزل الأس كمعامل
3. **Subtract 1 from the exponent**
   اطرح 1 من الأس
4. **Check: roots and fractions are powers too**
   √ والكسور قوى أيضًا

**Watch out**

- Subtract 1 from the exponent, do not divide by it.
- A negative exponent stays negative: −4 − 1 = −5.

**Tool.** d/dx x^r = r x^(r−1) · √x = x^(1/2) · 1/x = x^(−1)

## Worked example Authored (shape of Theorem 3.2 and Example 3.1)

**Use the General Power Rule to find $\frac{d}{dx}x^{-4}$ and $\frac{d}{dx}x^{1/3}$.**

1. bring −4 down, subtract 1 from the exponent

$$r=-4: \quad \frac{d}{dx}x^{-4}=-4x^{-4-1}$$

2. simplify the exponent

$$=-4x^{-5}$$

3. bring 1/3 down

$$r=\tfrac{1}{3}: \quad \frac{d}{dx}x^{1/3}=\tfrac{1}{3}x^{\frac{1}{3}-1}$$

4. 1/3 − 1 = −2/3

$$=\tfrac{1}{3}x^{-2/3}$$

**Your turn** · solve in your notebook · solutions at the back

1. Authored (shape of Theorem 3.2) Find $\frac{d}{dx}x^{3/4}$.
2. Authored (shape of Theorem 3.2) Find $\frac{d}{dx}x^{-5/3}$.
3. Authored (shape of Theorem 3.2) Find $\frac{d}{dx}x^{\sqrt{2}}$.

# Solutions

written out step by step · الحلول

## Objective 1 · Average and instantaneous velocity

السرعة المتوسطة واللحظية

### 1.1 Authored (shape of Ex 19)

$s(t) = 4t^2 + 3$ **feet. Find the average velocity between** $t = 1$ **and** $t = 3$**.**

1. substitute both times

$$s(3) = 36 + 3 = 39, \qquad s(1) = 4 + 3 = 7$$

2. change in position ÷ change in time

$$\frac{39 - 7}{3 - 1} = 16 \text{ ft/s}$$

### 1.2 Authored (shape of Ex 20)

$s(t) = t^3 - t$ **feet. Find the average velocity between** $t = 1.9$ **and** $t = 2$**, and estimate the instantaneous velocity at** $t = 2$**.**

1. substitute $t = 2$ and $t = 1.9$

$$s(2) = 8 - 2 = 6, \qquad s(1.9) = 6.859 - 1.9 = 4.959$$

2. the average on the short interval

$$\frac{6 - 4.959}{0.1} = 10.41 \text{ ft/s}$$

3. shorter and shorter intervals approach 11

the averages approach 11 : $v(2) \approx 11$ ft/s

### 1.3 Authored (shape of Ex 22)

$s(t) = 5\sin(t - 1)$ **feet (radians). Find the average velocity between** $t = 0.99$ **and** $t = 1$**, and estimate the instantaneous velocity at** $t = 1$**.**

1. calculator in radians

$$s(1) = 5\sin 0 = 0, \qquad s(0.99) = 5\sin(-0.01) \approx -0.05$$

2. average velocity on $[0.99, 1]$

$$\frac{0 - (-0.05)}{0.01} \approx 5.00 \text{ ft/s}$$

3. the averages approach 5

$v(1) \approx 5$ ft/s

## Objective 2 · The graph of the derivative

منحنى المشتقة

### 2.1 Authored (shape of Ex 13)

**The graph of $f(x) = x^2 - 4x + 3$ has its lowest point at $(2, -1)$. Describe the graph of $f'$: where is it zero, positive and negative, and what kind of curve is it?**

1. horizontal tangent at the vertex

   $f'(2) = 0$ (the lowest point)

2. sign of the slope on each side

   $f$ falls for $x < 2$ : $f' < 0$; rises for $x > 2$ : $f' > 0$

3. the slope grows steadily: a line (f'(x) = 2x − 4)

   $f'$ is a straight line through $(0, -4), (2, 0), (4, 4)$

### 2.2 Authored (shape of Ex 14)

**The graph of $f(x) = -x^3 + 3x$ has a high point at $x = 1$ and a low point at $x = -1$. Sketch $f'$: where is it zero, positive and negative?**

1. level points of f

   $f'(-1) = f'(1) = 0$

2. rising between the dip and the peak

   $f$ rises on $(-1, 1)$ : $f' > 0$; falls elsewhere: $f' < 0$

3. f'(x) = −3x² + 3

   $f'$ is a downward parabola, highest at $(0, 3)$

### 2.3 Authored (shape of Ex 15)

**The graph of $f(x) = \frac{1}{3}x^3 - x$ levels out at $x = -1$ and $x = 1$. Describe the graph of $f'$.**

1. level points of f

   $f'(-1) = f'(1) = 0$

2. f rises outside, falls between

   $f' > 0$ for $|x| > 1$, $f' < 0$ on $(-1, 1)$

3. f'(x) = x² − 1

   $f'$ is an upward parabola, lowest at $(0, -1)$

**3.1** Authored (shape of Theorem 3.2)

**Find** $\frac{d}{dx}x^{3/4}$.

1. bring 3/4 down and subtract 1

$\frac{3}{4}x^{\frac{3}{4}-1}$

2. 3/4 − 1 = −1/4

$= \frac{3}{4}x^{-1/4}$

**3.2** Authored (shape of Theorem 3.2)

**Find** $\frac{d}{dx}x^{-5/3}$.

1. bring −5/3 down and subtract 1

$-\frac{5}{3}x^{-\frac{5}{3}-1}$

2. −5/3 − 1 = −8/3

$= -\frac{5}{3}x^{-8/3}$

**3.3** Authored (shape of Theorem 3.2)

**Find** $\frac{d}{dx}x^{\sqrt{2}}$.

1. the rule works for any real exponent, irrational ones included

$\sqrt{2}\,x^{\sqrt{2}-1}$

# Ready?

Tick each line · ضع علامة

- [ ] **I can find average velocity and estimate instantaneous velocity**
  أستطيع إيجاد السرعة المتوسطة وتقدير اللحظية
- [ ] **I can sketch the graph of f′ from the graph of f**
  أستطيع رسم منحنى f′ من منحنى f
- [ ] **I can use the General Power Rule for any real exponent**
  أستطيع استخدام قاعدة القوة العامة

### On the day

Write the first line at once: the first step already earns a mark. In the sketch, mark the level points of f first.

### Every question

Show the two times and the division for a velocity; write the exponent subtraction for a derivative.

## One mark for every written step

لكل خطوة تكتبها علامة